

# Resource Allocation for URLLC/mMTC in 6G Industrial IoT Networks: Joint Optimization under Mixed-Numerology and SNR Constraints

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**Abstract**—As 6G smart factories accommodate a massive number of diverse devices, co-scheduling Ultra-Reliable Low-Latency Communications (URLLC) and Massive Machine-Type Communications (mMTC) services on shared physical resources emerges as a critical challenge. This paper proposes a resource allocation mechanism for this convergence scenario, combining SNR boosting strategy, flexible numerology configuration and priority control to device. We design a low-complexity heuristic scheduling algorithm. Simulation results indicate that our proposed method significantly enhances resource utilization while ensuring URLLC reliability, improving system schedulability by approximately 20.2%. This study delivers effective technical support for establishing industrial wireless communication systems capable of coexisting multi-QoS services.

**Index Terms**—6G Industrial IoT (IIoT), mixed numerology, URLLC/mMTC, resource allocation, heuristic QoS-aware scheduling.

## I. INTRODUCTION

With the rise of Industry 4.0, smart factories are transitioning from wired to wireless communication, supporting large-scale deployment of heterogeneous devices with diverse Quality of Service (QoS) requirements. These devices exhibit diverse communication requirements: devices like environmental sensors align with mMTC due to their tolerance to delay, while real-time control units, such as robotic arms, require stringent latency and reliability, demanding URLLC-level support [1]–[3].

These hybrid service demands challenge traditional cellular architectures. As outlined in the International Telecommunication Union (ITU) International Mobile Telecommunications (IMT)-2030 vision, sixth-generation (6G) aims to deliver  $< 0.1$  ms latency,  $10^{-7}$  reliability, and ultra-high device density [4], making efficient, flexible, and QoS-aware resource allocation critical [5].

Recent research has advanced resource allocation for URLLC in smart factories. One study [6] proposed a downlink URLLC model under the 5G New Radio (5G NR) standard, using a matching game and water level algorithm to jointly optimize time-frequency and power allocation, improving throughput. Building on this, another work [7] developed a dynamic ORAN-based framework that combines a transformer network with a Mixed-Integer Linear Programming (MILP)

model for real-time, QoS-aware scheduling, highlighting the benefits of flexible numerology in low-latency scenarios. For mMTC, a semi-persistent scheduler using multi-numerology and clustering [8] was introduced to enhance spectrum reuse and connection density with minimal signaling overhead, showing strong 6G scalability. Additionally, flexible multi-numerology systems featuring configurable subcarrier spacing, symbol durations, and subframe structures, have emerged as key enablers of diverse industrial applications in the 6G era [9], [10].

Despite recent advances, current research has several key limitations. First, most flexible numerology approaches assume idealized channel models, neglecting industrial wireless complexities like obstructions, multipath effects, and high-density deployments [7], [8], which limits practical applicability. Second, many URLLC resource allocation methods rely on fixed numerology and lack integration with flexible-numerology designs that adapt to dynamic QoS constraints [6]. Third, existing studies typically focus on either URLLC or mMTC in isolation, offering limited solutions for coexistence and conflict resolution within shared resource pools [6], [8], [10]. These gaps hinder effective support for heterogeneous service concurrency in 6G smart factory environments.

This paper proposes a QoS-focused resource allocation mechanism for 6G smart factories, optimizing collaboration between URLLC and mMTC devices with shared resources. The main objectives of this work can be summarized as follows:

- 1) We propose a novel Signal-to-Noise Ratio (SNR) boosting strategy that selectively enhances the transmit power for critical URLLC devices under non-line-of-sight (NLoS) conditions. This mechanism enhances the capacity to schedule more high-priority devices and ensures reliability under quasi-static channel environments.
- 2) We formulate a multi-objective resource allocation problem for 6G industrial networks that support both URLLC and mMTC devices, aiming to minimize system bandwidth usage and average communication latency under strict QoS delay constraints. To solve this, we design a scalable context-aware scheduling framework that integrates SNR-based filtering, numerology mapping, and

priority-driven allocation, effectively adapting to heterogeneous service requirements and dynamic channel conditions.

- 3) We evaluate the performance of our framework through simulation, demonstrating that our SNR-enhanced scheduling model improves URLLC reliability, reduces average latency, and achieves higher bandwidth efficiency. Results reveal a 20.2% improvement in system *schedulability*, enhancing resource utilization while maintaining URLLC reliability.

The paper is structured as follows. The scenario and system model are established in Section II, where channel modeling and the flexible time–frequency structure are presented. The problem formulation and the proposed scheduling algorithm are presented in Section III. Section V presents the performance set-up and discuss the results. The conclusion is given in Section VI.

## II. SYSTEM MODEL

### A. System Architecture

We consider a typical Industrial Internet of Things (IIoT) scenario with a single fifth-generation (5G) base station (gNodeB (gNB)) managing resource allocation for heterogeneous devices in a  $50 \times 50 \text{ m}^2$  factory environment. As illustrated in Fig. 1, the gNB is positioned 10 meters outside the factory to ensure full coverage [11]. A total of  $I = 100$  devices,

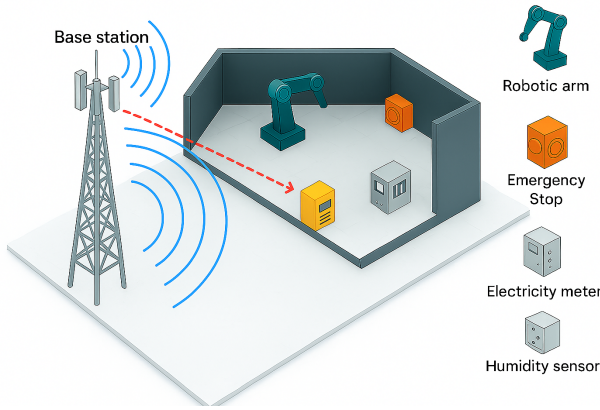


Fig. 1: Scenario and System Model

denoted by the set  $\{i_1, i_2, \dots, i_N\}$ , are randomly distributed and equipped with single antennas at height  $h_d = 1.5 \text{ m}$ . Devices connect to the base station for downlink control and data, while uplink and inter-cell interference are excluded due to the factory's isolated nature.

Devices are divided into two QoS classes:

- **URLLC devices** ( $I_{\text{URLLC}}$ ): High-priority devices such as robotic arms and emergency systems, requiring ultra-low latency and high reliability.
- **mMTC devices** ( $I_{\text{mMTC}} = I - I_{\text{URLLC}}$ ): Low-power sensors like meters and humidity monitors [3], characterized by looser latency demands but large-scale connectivity.

### B. Channel Model

This work presents a channel model for communication between static industrial devices in suburban factory settings, based on the Indoor Factory with Sparse clutter and Low base station height (InF-SL) scenario as specified in 3GPP TR 38.901 with a center frequency  $f_c = 3.5 \text{ GHz}$  [12], [13]. The model features sparsely distributed obstacles within the factory, such as machinery, and equipment, with terminal antenna heights lower than the average clutter height and a base station [12]. Due to the fixed equipment and base station, we adopt a quasi-static channel model [6]. This model ignores the effects of small-scale fast fading and considers only large-scale fading, which includes path loss and lognormal shadow fading.

1) **Line-of-sight (LoS) Probability Model:** In this context, 3GPP specifies a distance-dependent LoS probability model. Let  $d_{2D}$  represent the distance between the base station gNB and the device  $i_n$ . The probability of the device being in the LoS state at this distance can be expressed in an exponentially decaying form [12]:

$$\Pr_{\text{LoS,subsc}}(d_{2D}) = \exp\left(-\frac{d_{2D}}{k_{\text{subsc}}}\right) \quad (1)$$

Where:

$$k_{\text{subsc}} = \frac{-d_{\text{clutter}}}{\ln(1-r)} \quad (2)$$

The parameter  $k_{\text{subsc}}$  is the environmental attenuation distance, which is determined by the plant layout parameters, derived from the average obstacle spacing  $d_{\text{clutter}}$  and the empirical value  $r$ .

2) **Propagation Distance Modeling:** In accordance with the 3GPP model, we separate the total propagation distance into two segments in this paper: the outdoor and indoor parts, given the existence of a propagation path from outdoor to indoor. The total 3D propagation distance  $d_{3D,\text{total}}$  is calculated as [12]:

$$d_{3D,\text{total}} = \sqrt{(d_{2D,\text{out}} + d_{2D,\text{in}})^2 + (h_{\text{BS}} - h_{\text{UT}})^2} \quad (3)$$

where  $d_{2D,\text{out}}$  and  $d_{2D,\text{in}}$  are the horizontal distances from the base station to the exterior wall of the building and the horizontal distances of the signal propagation in the interior, respectively.  $h_{\text{BS}}$  and  $h_{\text{UT}}$  are the heights of the base station and antenna, respectively. The formula combines outdoor and indoor transmission paths into a single 3D propagation distance for path loss calculation, which is detailed in the next subsection.

3) **Path Loss (PL) Models:** In industrial scenarios such as InF-SL, different PL models are required for different propagation conditions. For the LoS condition, the path loss is usually modelled as the logarithm of distance and center frequency. Therefore, the LoS path loss model for InF-SL environment at center frequency  $f_c$  (GHz) can be expressed as [12]:

$$PL_{\text{LoS}} = 31.84 + 21.50 \log_{10}(d_{3D}) + 19.00 \log_{10}(f_c) \quad (4)$$

However, in the NLoS scenario, there is a high density of clutter inside the plant, and much of the equipment is obscured by objects such as shelves and walls. Therefore, the NLoS path loss is modelled as [12]:

$$PL_{\text{NLoS}}^{\text{InF-SL}} = 33 + 25.5 \log_{10}(d_{3D}) + 20 \log_{10}(f_c) \quad (5)$$

$$PL_{\text{NLoS}} = \max(PL, PL_{\text{LoS}}) \quad (6)$$

The NLoS scenario exhibits a higher distance attenuation factor, indicating significant added attenuation in densely cluttered settings.

4) **SNR Calculation:** Based on the above large-scale fading channel model, the SNR of each device is obtained by modelling using the discriminative mechanism  $\text{Pr}_{\text{LoS,subsc}}$ .

The SNR of a single link can be determined from the transceiver parameters and path loss. The base station transmitter power is denoted as  $P_{\text{tx}}$ , the transmit and receive antenna gains are  $G_{\text{tx}}$  and  $G_{\text{rx}}$ , respectively, and the path loss after inclusion of shadowing fading is  $PL$ . The receiver noise power is  $N_0$ , which can be calculated from the Boltzmann constant [14]. The SNR is defined as the ratio of the signal power to the noise power. Therefore, the SNR is given by:

$$\text{SNR}_i = P_{\text{tx}} + G_{\text{tx}} + G_{\text{rx}} - PL - N_0 \quad (7)$$

The primary role of the SNR in this work is to measure each device's wireless link quality, acting as a crucial factor in deciding *schedulability*.

### C. Flexible Time—frequency resource structure in 5G NR

To accommodate diverse QoS needs, especially for URLLC in industrial settings, all devices and the base station utilize a complete bandwidth of  $B_{\text{tot}} = 100$  MHz within a licensed 6G industrial spectrum [13].

5G NR introduces a flexible numerology framework, where each numerology corresponds to a specific Subcarrier Spacing (SCS) and symbol duration [8]. This study considers three configurations:  $\Delta f = 15, 30$ , and  $60$  kHz, enabling adaptable transmission intervals from milliseconds to sub-milliseconds, well-suited for heterogeneous traffic.

To meet the stringent latency demands of URLLC, a two-symbol mini-slot structure is adopted for  $\Delta f = 60$  kHz, with a duration of  $16.67 \mu\text{s}$ , capable of completing downlink transmissions within the  $1$  ms latency constraint [15]. For more delay-tolerant services, lower SCS values ( $15$  or  $30$  kHz) improve spectral efficiency.

We consider the resource allocation framework by defining a Resource Block (RB) as the fundamental scheduling unit, representing a set number of subcarriers across the frequency domain and varying mini-slot or slot durations based on time

domain numerology. For example, an RB at  $15$  kHz covers  $12$  subcarriers  $\times 0.5$  ms, while at  $60$  kHz, it covers  $12$  subcarriers  $\times 0.25$  ms. Let  $\mathcal{R}$  signify the total available RBs within each  $1$  ms scheduling interval. The RB count in  $\mathcal{R}$  is determined by the chosen numerology configuration and total bandwidth  $B_{\text{tot}}$ , forming the entire resource pool for downlink scheduling.

### III. PROBLEM FORMULATION

We tackle the cooperative scheduling challenges of URLLC and mMTC devices with limited wireless resources by creating a multi-objective optimization framework. Extending the approach in [8], this joint optimization not only minimizes total bandwidth usage but also incorporates average communication latency per device, ensuring that each device's QoS requirements are met.

To balance the trade-off between bandwidth efficiency and latency performance, we formulate a weighted multi-objective optimization. Let  $\alpha \in [0, 1]$  and  $\beta = 1 - \alpha$  represent weighting factors for optimizing resource utilization and latency, respectively. Both latency and bandwidth usage are normalized to ensure fairness and prevent one objective from dominating the other. The overall problem is mathematically defined as optimization problem  $\mathcal{P}$ .

$$\min \quad \alpha \sum_{n=1}^{n_{\max}} z_n f_n + \beta L_{\text{avg}} \quad (8)$$

$$\text{s.t.} \quad \sum_{n=1}^{n_{\max}} x_{in} = 1, \quad \forall i \in \mathcal{I} \quad (8a)$$

$$\sum_{i \in \mathcal{I}} x_{in} \leq |\mathcal{I}| z_n, \quad \forall n \in [1, n_{\max}] \quad (8b)$$

$$\tau_n + \sum_{i \in \mathcal{H}_i} [p_i/p_1] \tau_n \leq L_i, \quad \forall i \in \mathcal{I}, \forall n \in [1, n_{\max}] \quad (8c)$$

$$\{x_{in}, p_i, j_i, f_n\}_{i \in \mathcal{I}} \in \mathcal{S}', \quad \forall n \in [1, n_{\max}] \quad (8d)$$

$$\text{SNR}_i \geq \text{SNR}_{\text{threshold}}, \quad \forall i \in \mathcal{I}_{\text{candidate}} \quad (8e)$$

The objective function in Equation 8 minimizes total bandwidth usage and average latency across the scheduled device set  $\mathcal{I}$ . Bandwidth usage is measured by the number of RBs allocated, where each device is assigned one RB from a set of bands  $n \in \{1, \dots, n_{\max}\}$ , each linked to a numerology type  $\mu \in \{0, 1, 2\}$ . A binary variable  $z_n$  indicates whether any mMTC device uses band  $n$ , and  $f_n$  represents the per-RB bandwidth, selected from  $\{f^1, f^2, \dots, f^{|k|}\}$ . The second component,  $L_{\text{avg}}$ , captures the average latency across scheduled devices, incorporating both transmission and queuing delays. This term is optimized to meet the stringent QoS requirements of URLLC services, which can be defined as:

$$L_{\text{avg}} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \sum_{n=1}^{n_{\max}} x_{in} \cdot \left( \tau_n + \sum_{l \in \mathcal{H}_i} \left[ \frac{p_i}{p_l} \right] \tau_n \right) \quad (9)$$

where  $|\mathcal{I}|$  denotes the total number of devices that have successfully received resource allocation in device  $\mathcal{I}$ . Each device

TABLE I: 5G NR Numerology Parameters and Applicable Scenarios

| Numerology ( $\mu$ ) | SCS (kHz) | Slot Duration (ms) | Cyclic Prefix | Symbol Duration ( $\mu$ s) | Applicable Scenarios |
|----------------------|-----------|--------------------|---------------|----------------------------|----------------------|
| 0                    | 15        | 1.0                | Normal        | $\approx 66.67$            | eMBB, mMTC           |
| 1                    | 30        | 0.5                | Normal        | $\approx 33.33$            | mMTC, moderate URLLC |
| 2                    | 60        | 0.25               | Normal        | $\approx 16.67$            | URLLC, low latency   |

$i_n \in \mathcal{I}$  is characterized by a fixed packet generation period  $p_i$ .  $\mathcal{H}_i$  denotes the set of devices with higher priority than device  $i$ , and  $\tau_n$  represents the latency for data transmission. And  $p_l$  is the packet generation period of  $\mathcal{H}_i$ , which is used to calculate the queueing delay.

To ensure practical applicability, the optimization problem is subject to a set of constraints. Constraint 8a specifies that a device can only be assigned to one band.  $x_{in}$  is a binary variable, if  $x_{in} = 1$ , it means device  $i$  has been assigned to band  $n$ , so it cannot be assigned again. Constraint 8b states that if at least one device is assigned to a band, the band must be enabled. Moreover, the total latency of a device  $L_{\text{total}}$ , comprises queueing delay and transmission delay. Therefore, constraint 8c limits the total latency of each assigned device  $i$  ensuring it does not exceed the maximum latency requirement  $L_i$  of its QoS class. Considering the actual quality of wireless channel, constraint 8e ensures channel condition by requiring the SNR of all scheduled devices to meet a predetermined threshold  $\text{SNR}_{\text{threshold}}$ . To ensure the delay sensitivity of critical services during resource allocation, we propose the priority service rule as an implicit constraint. This rule mandates that URLLC devices (Priority=1) are serviced prior to mMTC devices (Priority=0).

This issue falls under the category of a mixed-integer nonlinear programming (MINLP) because it involves discrete binary variables,  $z_n$  and  $x_{in}$ , alongside nonlinear latency calculations. Recognized as NP-hard, MINLP problems do not have a polynomial-time optimal solution. Thus, this paper demonstrates the effectiveness of the proposed algorithm by verifying that both latency and bandwidth requirements meet QoS requirements.

#### IV. SOLUTION TO THE MULTI-OBJECTIVE OPTIMIZATION PROBLEM

The proposed algorithm addresses the NP-hard problem of binary scheduling variables and nonlinear delay constraints by integrating service prioritization and channel quality-aware resource allocation. Unlike existing work [8], our approach introduces a novel SNR enhancement mechanism and multi-numerology mapping strategy for low-latency, high-reliability URLLC services, while maximizing bandwidth efficiency and maintaining fairness for mMTC devices.

##### A. Initialization Phase

During the initialization phase, devices are classified and wireless resources are prepared for subsequent SNR estimation and resource allocation. The system iterates through available frequency bands, initializing the total number of resource blocks  $\text{Total\_RB}[n]$ , and resetting both the used

resource blocks  $\text{RB\_Used}[n]$  and the consumed bandwidth  $\text{UsedBW}[n]$  to zero. Devices are then prioritized based on service type: URLLC devices are assigned Priority = 1, while other devices are marked with Priority = 0, which influences their scheduling order. Finally, channel conditions are assessed by estimating the line-of-sight (LoS) probability  $\text{Pr}_{\text{LoS,subsec}}$  for each device. The result is stored as  $\text{deviceLOS}[i]$ , guiding subsequent SNR enhancement and scheduling decisions.

##### B. SNR Estimation and Boosting Phase

As illustrated in Algorithm 1, an accurate assessment of device wireless channel quality is performed prior to resource allocation to enhance resource utilization and ensure communication reliability. We begin by calculating the SNR of each device in the sets  $\text{deviceList}_{\text{URLLC}}$  and  $\text{DeviceList}_{\text{mMTC}}$  using the initial transmit power  $P_{\text{tx\_global}}$ , filtering those that exceed the threshold  $\text{SNR}_{\text{threshold}}$  (Line 3). To improve the reliability of mission-critical communications, a transmit power boosting strategy is applied: URLLC devices in NLoS conditions with sub-threshold SNR values are assigned a higher power level  $P_{\text{tx\_boosted}}$  (Line 4). The SNR is then recalculated (Line 5), and devices that now meet or exceed  $\text{SNR}_{\text{threshold}}$  are included in the candidate set  $\mathcal{I}_{\text{candidate}}$  for scheduling and optimization (Line 6).

##### Algorithm 1 SNR Estimation and Boosting Phase - Joint URLLC/mMTC Resource Allocation

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**Input:**  $\text{DeviceList}_{\text{URLLC}}$ ,  $\text{DeviceList}_{\text{mMTC}}$ , initial transmit power  $P_{\text{tx\_global}}$ ,  $\text{SNR}_{\text{threshold}}$

**Output:**  $\mathcal{I}_{\text{candidate}}$

```

1: for each device  $i \in \mathcal{I}$  do
2:   Compute initial SNR:  $\text{SNR}[i] \leftarrow \text{compute\_SNR}(P_{\text{tx\_device}}[i], i)$ 
3:   if  $\text{deviceLOS}[i] = \text{NLOS} \wedge \text{Priority}[i] = 1 \wedge \text{SNR}[i] < \text{SNR}_{\text{threshold}}$  then
4:     Boost transmit power:  $P_{\text{tx\_boosted}} \leftarrow P_{\text{tx\_global}} + 10 \text{ dB}$ 
5:     Recompute SNR:  $\text{SNR}[i] \leftarrow \text{compute\_SNR}(P_{\text{tx\_boosted}}, i)$ 
6:   end if
7:   Identify eligible device set:  $\mathcal{I}_{\text{candidate}} \leftarrow \{i \mid \text{SNR}[i] \geq \text{SNR}_{\text{threshold}}\}$ 
8: end for
    
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##### C. Scheduling Phase

After SNR estimation and forming the candidate device set  $\mathcal{I}_{\text{candidate}}$ , the resource allocation phase begins, as detailed in Algorithm 2. There are two stages in Algorithm 2, stage 1 is for URLLC device and stage 2 is for mMTC device. Resources are distributed according to device priority and QoS categories, prioritizing URLLC services for low-latency, high-reliability



needs. Thus, URLLC devices receive resource allocation first (Stage 1).

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**Algorithm 2** Scheduling Phase - Joint URLLC/mMTC Resource Allocation

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**Input:**  $\mathcal{I}_{\text{candidate}}$ ,  $\text{DeviceList}_{\text{URLLC}}$ ,  $\text{DeviceList}_{\text{mMTC}}$ , QoS classes, initial transmit power  $P_{\text{tx\_global}}$ , SNR threshold

**Output:** Resource allocation results (scheduled devices, RB utilization)

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**Stage 1:** URLLC Device Scheduling (Priority = 1)

```

1: Sort  $\text{DeviceList}_{\text{URLLC}}$  by increasing  $L_i \triangleright$  Delay-sensitive first
2: for each device  $i \in \text{DeviceList}_{\text{URLLC}}$  do
3:   Get QoS class  $m \leftarrow \text{QoS\_Class}[i]$ 
4:   for each numerology  $\mu \in \text{MappingTable}[m]$  do
5:     if  $RB_{\text{Available}}[\mu] \geq RB_i$  then
6:       Allocate  $RB_i$  in numerology  $\mu$  to device  $i$ 
7:        $RB_{\text{Available}}[\mu] \leftarrow RB_{\text{Available}}[\mu] - RB_i$ 
8:        $RB_{\text{Used}}[\mu] \leftarrow RB_{\text{Used}}[\mu] + RB_i$ 
9:       Mark  $i$  as allocated
10:    break
11:   end if
12: end for
13: end for

```

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Initially, We prioritize URLLC devices in  $\text{DeviceList}_{\text{URLLC}}$  meeting  $\text{SNR}_{\text{threshold}}$ , sorting them by latency requirements  $L_i$  in ascending order to address latency-sensitive devices first. Each URLLC device is then classified into one of  $m$  QoS classes based on latency, mapped to a preferred numerology  $\mu$  based on a mapping table. During resource allocation, the base station prioritizes these numerologies accordingly (Lines 2–4). If RBs remain available across all numerologies, the base station allocates one RB to device  $i$  and updates the available resources. If the device is marked as ‘Scheduled’, the current scheduling loop for it is exited (Lines 5–10). This phase ensures URLLC devices are prioritized in bandwidth-constrained scenarios, demonstrating the algorithm’s delay-aware design.

We continue to allocate mMTC devices in stage 2 with the available numerology resources. The scheduling process is structurally identical to that of URLLC, following the same QoS-based numerology mapping. To avoid redundancy, the detailed steps are omitted.

## V. SIMULATION RESULTS

In this section, we present the simulation results of the Joint URLLC/mMTC Resource Allocation algorithm with the optimization objective described in section III.

### A. Simulation Setup

Simulations were conducted using a MATLAB-based system-level model of a  $50\text{m} \times 50\text{m}$  industrial factory. A base station was placed 35 m outside the factory center at a height of 5 m, with devices on the ground. The system operated at 3.5 GHz with a 100 MHz bandwidth and 30 dBm transmit

power. The UT noise figure was 9 dB, and antenna gains were  $-15\text{dBi}$  (BS) and  $3\text{dBi}$  (devices).

A total of 100 devices were randomly placed using Monte Carlo sampling, comprising 70% URLLC and 30% mMTC to reflect industrial service requirements. Three numerology options were supported: 15 kHz ( $\mu = 0$ ), 30 kHz ( $\mu = 1$ ), and 60 kHz ( $\mu = 2$ ). URLLC employed 2-symbol mini-slots for low latency, while mMTC used 14-symbol normal-slots for higher efficiency.

SNR was computed considering device location, path loss, and system parameters, with a scheduling threshold of 15 dB. Symbol durations, derived from subcarrier spacing, were  $66.7\text{ }\mu\text{s}$  (15 kHz),  $33.3\text{ }\mu\text{s}$  (30 kHz), and  $16.7\text{ }\mu\text{s}$  (60 kHz), defining slot structures for subsequent performance analysis.

### B. Result and Discussion

Figure 2 illustrates the SNR (dB) distribution of devices before and after enhancement in the initial Monte Carlo simulation, sorted by LoS and NLoS links. Initially (upper panel), some NLoS URLLC devices have SNRs below 15 dB (black dashed line), indicating reliability issues. To resolve this, we introduce a priority-based SNR enhancement mechanism. Devices with SNR below the threshold (Priority=1) receive an 8 dB transmit power boost ( $P_{\text{tx\_boosted}} = P_{\text{tx}} + 8$ ), consistent with medium BS power limits [12].

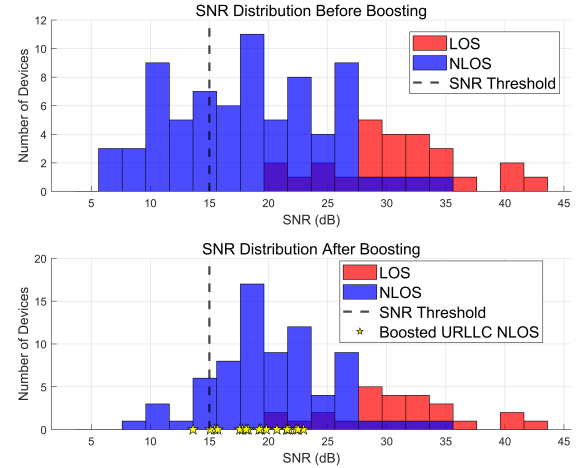


Fig. 2: SNR Distribution Before and after Boosting

Enhanced SNRs are recalculated considering PL, shadow fading, antenna gain, and noise in line with 6G requirements. In the post-enhancement scenario (lower panel), all NLoS URLLC devices (yellow stars) exceed the threshold, enabling resource allocation. This proactive boosting enhances reliability and addresses fading in edge devices by improving crucial channel qualities without adding system complexity.

Figure 3 demonstrates the delay distribution, bandwidth utilization, and resource allocation efficiency for URLLC devices before and after SNR enhancement. The top-left and top-right subfigures show the delay distributions in both scenarios.

Resource allocation uses mini-slot transmission and QoS–Numerology mapping, prioritizing delay-sensitive devices with 60 kHz numerology to reduce latency.

Before SNR enhancement, 51 devices met the 1 ms URLLC latency threshold with an average latency of 0.467 ms. After enhancement, 69 devices satisfied the threshold, with a slightly increased average latency of 0.594 ms, still within the URLLC requirement. This improvement results from SNR boosting, which enhances reception quality for devices in poor channels by increasing transmit power. Despite a slight increase in latency due to poorer channel conditions and higher scheduling demands, enhanced SNR compliance significantly improves service reachability and system capacity.

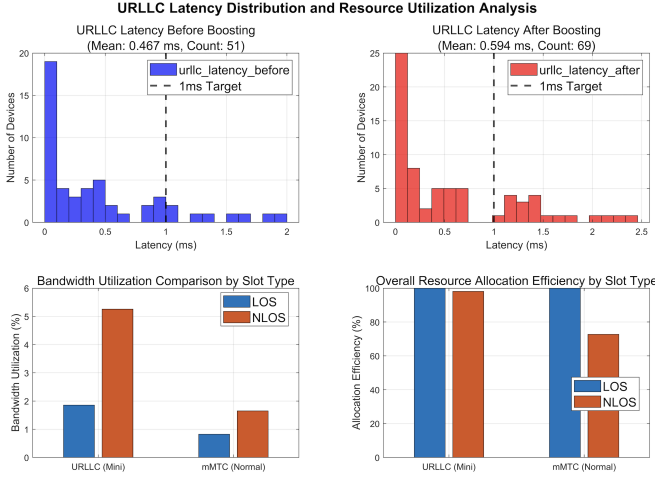


Fig. 3: URLLC Latency Distribution and Resource Utilization Analysis

The lower-left subfigure in Figure 3 compares bandwidth utilization under LoS/NLoS conditions for different service types. URLLC achieves 5.2% utilization under NLoS using the mini-slot scheme, significantly outperforming the approximately 2.0% utilization of mMTC’s normal-slot configuration. This reflects the algorithm’s effectiveness in prioritizing high-QoS services while conserving bandwidth.

The lower-right subfigure shows resource allocation efficiency, reaching nearly 100% for URLLC and around 74% for mMTC under both LoS and NLoS conditions. These outcomes demonstrate the hierarchical scheduler’s ability to meet QoS demands while minimizing spectrum waste. Overall, the results confirm the robustness and efficiency of the proposed algorithm in mixed channel conditions with diverse QoS requirements.

Figure 4 shows that, although the number of URLLC devices increased from 51 to 69 after SNR boosting and average latency rose from 0.467 ms to 0.594 ms, the system still maintains strong delay control. The rightward shift in the CDF curve is due to two main factors: (1) SNR boosting targets NLoS URLLC devices, which face higher path loss and shadow fading, requiring more resources to meet the SNR threshold; and (2) increased device count intensifies scheduling

competition, potentially deprioritizing high-SNR but lower-priority devices. Despite this shift, the figure confirms that the proposed scheduling approach effectively balances latency, SNR, and resource usage, making it well-suited for 6G industrial IoT scenarios requiring reliable and low-latency URLLC service.

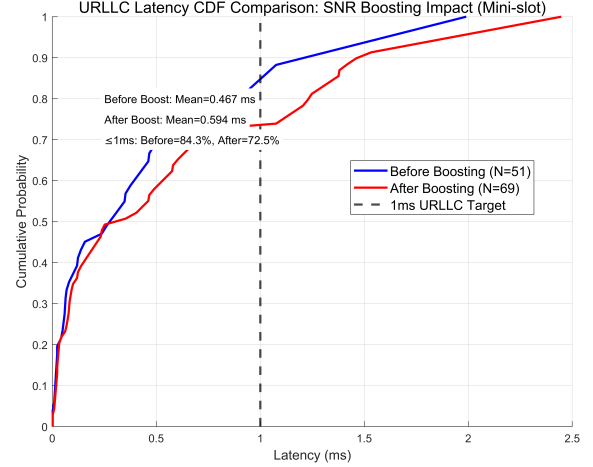


Fig. 4: URLLC Latency CDF Comparison: SNR Boosting Impact (Mini-slot)

In wireless systems, stochastic elements like device location, path loss, and shadow fading affect performance, making single-run simulations insufficient for comprehensive evaluation. To assess the robustness of the proposed scheduling and SNR enhancement algorithm, 40 independent Monte Carlo simulations were conducted, each with varying device distributions and path loss conditions. The top subplot of Figure 5 presents the number of additional URLLC devices included in each run due to SNR boosting, highlighting the algorithm’s consistency across diverse scenarios.

Results from 40 Monte Carlo runs show that SNR boosting added an average of 14.2 URLLC devices per run, with a standard deviation of  $\pm 3.3$ , reflecting a 20.2% improvement. This confirms the strategy’s effectiveness across diverse stochastic conditions. The middle subplot of Figure 5 shows that average URLLC latency increased from  $0.405 \pm 0.090$  ms to  $0.524 \pm 0.135$  ms post-boosting, still well within the 1 ms requirement, satisfying Industrial IoT QoS demands.

The bottom subplot compares resource utilization: URLLC occupied  $74.2\% \pm 2.2\%$  of system resources, while mMTC used  $25.8\% \pm 2.2\%$ , consistent with the 70% URLLC ratio and prioritization strategy. Low fluctuation across runs highlights stable and differentiated resource allocation.

Overall, this comparative analysis confirms that the proposed heuristic algorithm meets QoS requirements, improves bandwidth use, reduces delay, and maintains robustness under varying conditions, making it suitable for dynamic, multi-service industrial IoT scenarios.

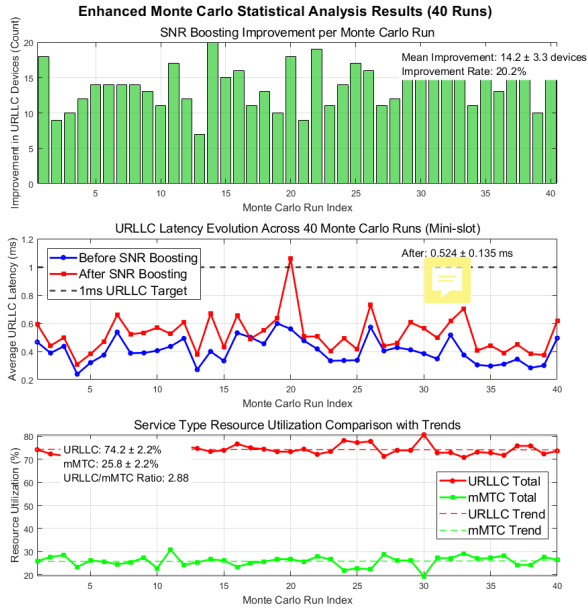


Fig. 5: Enhanced Monte Carlo Statistical Analysis Results (40 Runs)

## VI. CONCLUSION

This paper presents a novel resource allocation framework for coexisting URLLC and mMTC services in 6G smart factories, integrating flexible numerology and priority-aware scheduling at the physical layer. The proposed model jointly considers device types, delay constraints, and channel conditions, using heuristic algorithms for low-complexity optimization. A priority-based SNR enhancement strategy is introduced to improve the schedulability of mission-critical URLLC devices via transmit power boosting.

Simulation results show a  $\approx 20.2\%$  increase in the number of URLLC devices meeting scheduling requirements, with average latency maintained at 0.524 ms within industrial real-time limits. The algorithm achieves a robust balance between QoS prioritization and spectral efficiency, validating its applicability in heterogeneous industrial environments.

Future work will explore reinforcement learning for adaptive optimization and extend the framework to multi-node cooperative scheduling, enhancing scalability and system intelligence.

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